Jupyter Lab Usage

Jupyter Lab Usage

- 1. Jupyter Lab Installation
  - 1.1 Installing Dependencies
  - 1.2 Installing Jupyter Lab
- 2. Jupyter Lab Configuration
  - 2.1. Creating the Configuration File
  - 2.2. Modifying the Configuration File
  - 2.3. Configuring the Access Password
- 3. Launching Jupyter Lab
  - 3.1. Starting Jupyter Lab
  - 3.2. Host Access
- 4. Service Autostart at Boot
  - 4.1. Editing the Service File
  - 4.2. Setting Up the Autostart Service
  - 4.3. Verify Automatic Startup
- 5. Using JupyterLab
  - 5.1. Running Programs
  - 5.2. Kernel

1. Jupyter Lab Installation

1.1 Installing Dependencies

```
sudo pip3 install lark --ignore-installed --break-system-packages
sudo pip3 install jsonschema --ignore-installed --break-system-packages
sudo pip3 install ipywidgets --break-system-packages
```

```
jetson@ubuntu:~$ sudo pip3 install lark --ignore-installed --break-system-packag
es
Collecting lark
  Downloading lark-1.3.1-py3-none-any.whl.metadata (1.8 kB)
Downloading lark-1.3.1-py3-none-any.whl (113 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 113.2/113.2 kB 992.8 kB/s eta 0:00:00
Installing collected packages: lark
Successfully installed lark-1.3.1
```

```
jetson@ubuntu:~$ sudo pip3 install jsonschema --ignore-installed --break-system-
packages
Collecting jsonschema
  Downloading jsonschema-4.26.0-py3-none-any.whl.metadata (7.6 kB)
Collecting attrs>=22.2.0 (from jsonschema)
  Using cached attrs-25.4.0-py3-none-any.whl.metadata (10 kB)
Collecting jsonschema-specifications>=2023.03.6 (from jsonschema)
  Downloading jsonschema_specifications-2025.9.1-py3-none-any.whl.metadata (2.9
kB)
Collecting referencing>=0.28.4 (from jsonschema)
  Downloading referencing-0.37.0-py3-none-any.whl.metadata (2.8 kB)
Collecting rpds-py>=0.25.0 (from jsonschema)
```

```
jetson@ubuntu:~$ sudo pip3 install ipywidgets --break-system-packages
DEPRECATION: Loading egg at /usr/local/lib/python3.12/dist-packages/jetcam-0.0.0
-py3.12.egg is deprecated. pip 24.3 will enforce this behaviour change. A possib
le replacement is to use pip for package installation.. Discussion can be found
at https://github.com/pypa/pip/issues/12330
Collecting ipywidgets
  Downloading ipywidgets-8.1.8-py3-none-any.whl.metadata (2.4 kB)
Requirement already satisfied: comm>=0.1.3 in /usr/local/lib/python3.12/dist-pac
kages (from ipywidgets) (0.2.3)
Requirement already satisfied: ipython>=6.1.0 in /usr/local/lib/python3.12/dist-
packages (from ipywidgets) (9.11.0)
Requirement already satisfied: traitlets>=4.3.1 in /usr/local/lib/python3.12/dis
t-packages (from ipywidgets) (5.14.3)
Collecting widgetsnbextension~=4.0.14 (from ipywidgets)
  Downloading widgetsnbextension-4.0.15-py3-none-any.whl.metadata (1.6 kB)
```

1.2 Installing Jupyter Lab

Use the following command to install Jupyter Lab: If the download speed of Jupyter Lab is slow, you can use a specified source for installation.

```
sudo pip3 install jupyterlab --break-system-packages
```

```
# Tsinghua University source: sudo pip3 install jupyterlab --break-system-
packages -i https://pypi.tuna.tsinghua.edu.cn/simple
# Alibaba Cloud mirror: sudo pip3 install jupyterlab --break-system-packages -i
https://mirrors.aliyun.com/pypi/simple/
```

```
jetson@ubuntu:~$ sudo pip3 install jupyterlab --break-system-packages -i https:/
/pypi.tuna.tsinghua.edu.cn/simple
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Collecting jupyterlab
  Using cached https://pypi.tuna.tsinghua.edu.cn/packages/e1/1b/dad6fdcc658ed7af
26fdf3841e7394072c9549a8b896c381ab49dd11e2d9/jupyterlab-4.5.6-py3-none-any.whl (
12.4 MB)
Collecting async-lru>=1.0.0 (from jupyterlab)
  Using cached https://pypi.tuna.tsinghua.edu.cn/packages/13/5c/af990f019b8dd11c
5492a6371fe74a5b0276357370030b67254a87329944/async_lru-2.2.0-py3-none-any.whl (7
```

2. Jupyter Lab Configuration

Configure Jupyter Lab for local area network access, access password, and automatic startup.

Device on the same local area network can access it by entering IP:8888 in their browser!

Note: Accessing the local network (LAN) is often restricted on campus networks; you may need to switch to a laptop or mobile hotspot for testing purposes.

```
For example, if the board's IP address is 192.168.2.114, you can access the
board's Jupyter Lab interface by entering 192.168.2.114:8888 into a web browser
while connected to the same local network.
```

2.1. Creating the Configuration File

```
jupyter lab --generate-config
```

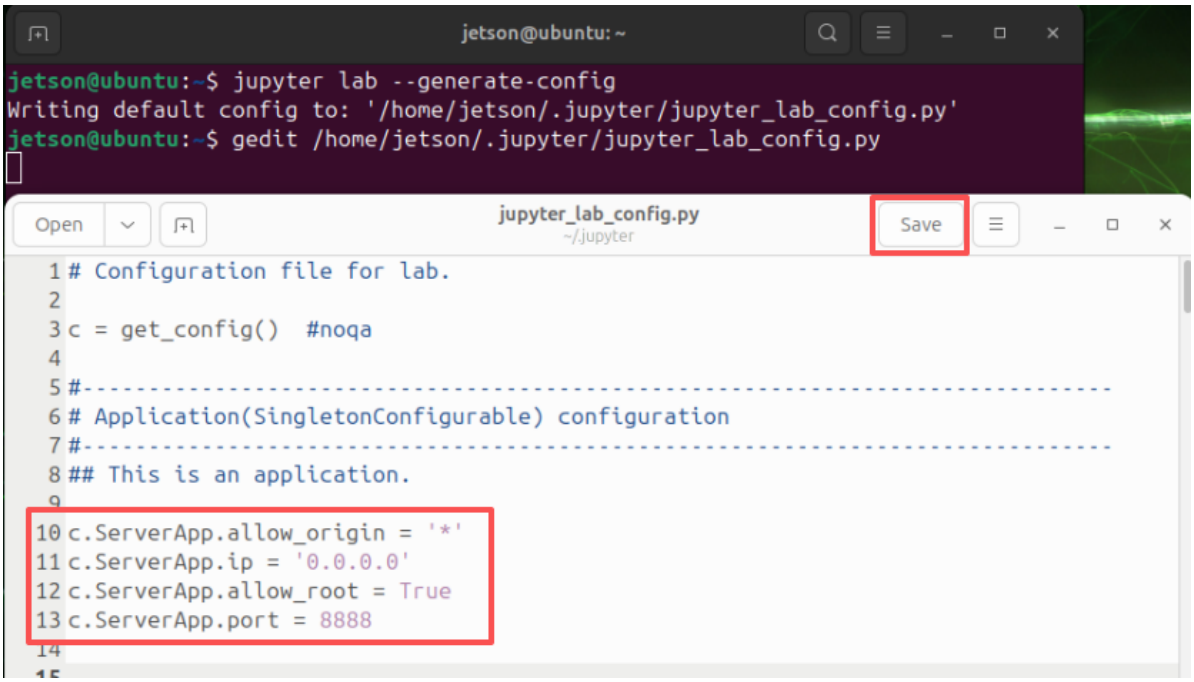
The automatically generated configuration file is located at: `/home/jetson/.jupyter/jupyter_lab_config.py`

2.2. Modifying the Configuration File

```
gedit /home/jetson/.jupyter/jupyter_lab_config.py
```

Add the following content to the file. After adding the content, click "Save" and close the file.

```
# Allow requests from any origin to access the Jupyter Lab server
c.ServerApp.allow_origin = '*'
# 0.0.0.0 binds to all available network interfaces, allowing access from any
address
c.ServerApp.ip = '0.0.0.0'
# Allow the Jupyter Lab server to be launched as the root user
c.ServerApp.allow_root = True
# Change the default port to avoid conflicts
c.ServerApp.port = 8888
```



2.3. Configuring the Access Password

Enter the password-setting command in the terminal. You will be prompted to enter the same password twice; for security reasons, the characters you type will not be displayed on the screen. Press the Enter key once you have finished typing!

Note: You must remember this password, as it will be required for future logins to Jupyter Lab. To make it easier to remember, the factory image has already been configured to use the same password as the host system: `yahboom`.

```
jupyter lab password
```

Location of the automatically generated configuration file: [JupyterPasswordApp] Wrote hashed password to /home/jetson/.jupyter/jupyter_server_config.json

```
jetson@ubuntu:~$ jupyter lab password
Enter password:
Verify password:
[JupyterPasswordApp] Wrote hashed password to /home/jetson/.jupyter/jupyter_serv
er_config.json
jetson@ubuntu:~$
```

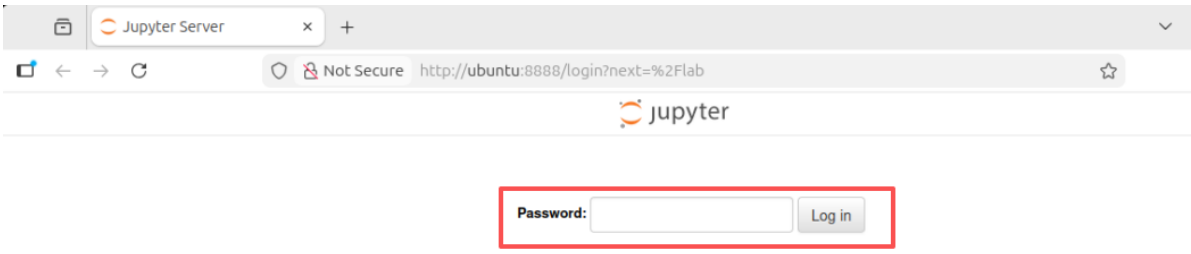
3. Launching Jupyter Lab

3.1. Starting Jupyter Lab

```
jupyter lab
# Launch without a browser: jupyter lab --no-browser
# Launch as administrator: sudo jupyter lab --allow-root
```

```
jetson@ubuntu:~$ jupyter lab
[I 2026-03-12 11:07:28.799 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-03-12 11:07:28.802 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-03-12 11:07:28.805 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-03-12 11:07:29.172 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-03-12 11:07:29.186 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-03-12 11:07:29.188 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-03-12 11:07:29.189 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-03-12 11:07:29.190 LabApp] JupyterLab extension loaded from /usr/local/lib/python3.12/dist-pack
ages/jupyterlab
[I 2026-03-12 11:07:29.191 LabApp] JupyterLab application directory is /usr/local/share/jupyter/lab
[I 2026-03-12 11:07:29.191 LabApp] Extension Manager is 'pypi'.
[I 2026-03-12 11:07:29.234 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-03-12 11:07:29.235 ServerApp] Serving notebooks from local directory: /home/jetson
[I 2026-03-12 11:07:29.235 ServerApp] Jupyter Server 2.17.0 is running at:
[I 2026-03-12 11:07:29.235 ServerApp] http://ubuntu:8888/lab
[I 2026-03-12 11:07:29.235 ServerApp]     http://127.0.0.1:8888/lab
[I 2026-03-12 11:07:29.235 ServerApp] Use Control-C to stop this server and shut down all kernels (twic
e to skip confirmation).
```

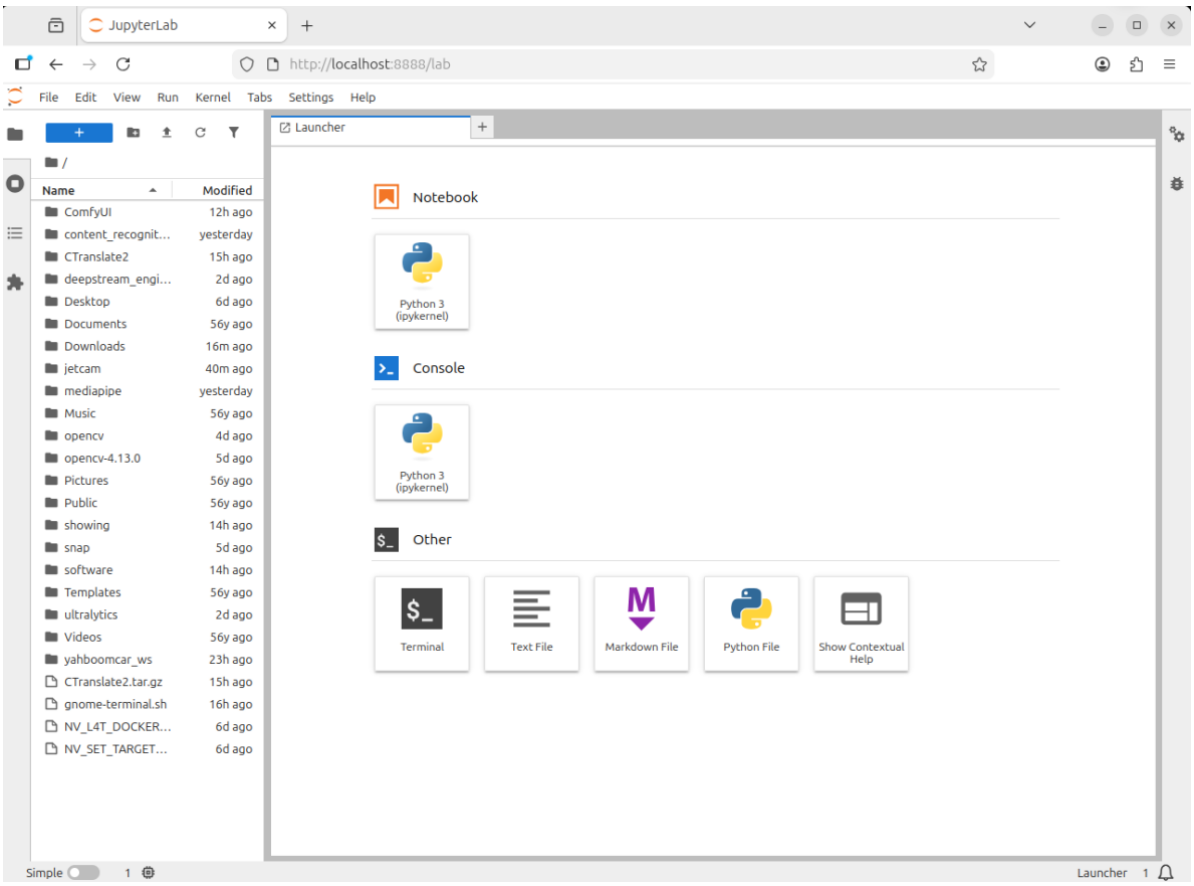
Upon first access, you will need to enter the password and then click [Log in] to proceed.



3.2. Host Access

"Host access" refers to accessing the service directly from the Jetson board's operating system. Open a web browser and enter `http://localhost:8888/` to access it:

```
http://localhost:8888/
```



4. Service Autostart at Boot

4.1. Editing the Service File

```
sudo gedit /etc/systemd/system/jupyterlab.service
```

Add the following content: After adding it, click "Save" and close the file.

```
[Unit]
Description=jupyterlab service
After=network.target
[Service]
Type=simple
ExecStart=/usr/local/bin/jupyter-lab
config=/home/jetson/.jupyter/jupyter_lab_config.py --no-browser
User=jetson
Group=jetson
WorkingDirectory=/home/jetson/
Restart=always
RestartSec=10
[Install]
```

```
WantedBy=multi-user.target
```

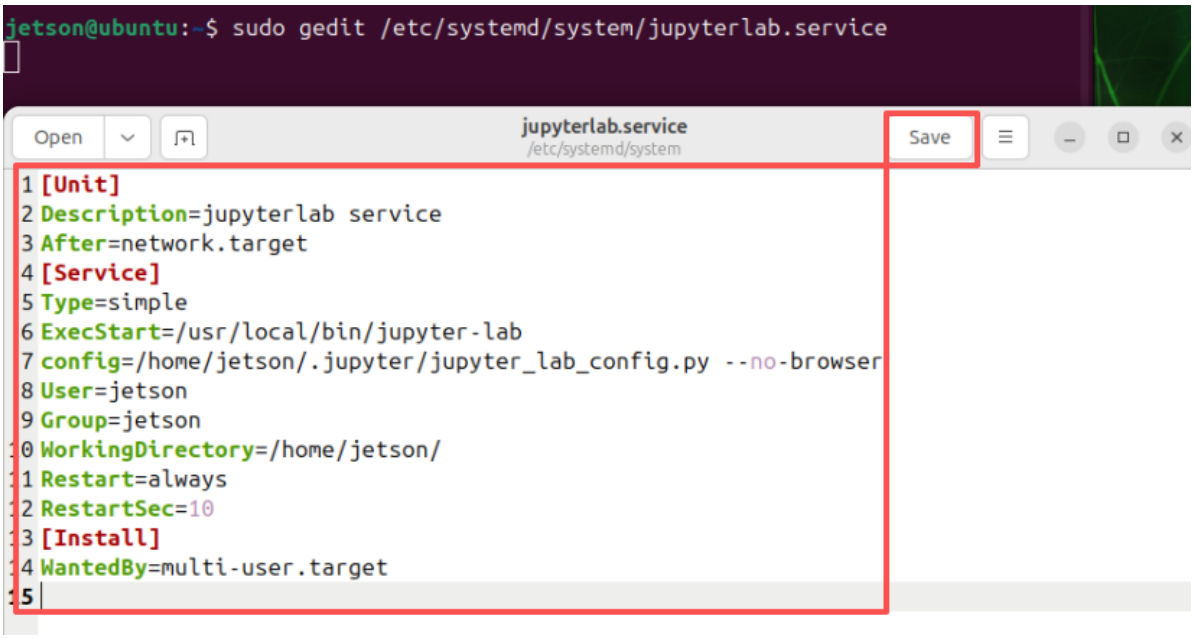
`jetson`: The username.

`ExecStart`: The command used to launch JupyterLab; modify this to match the actual installation path of JupyterLab.

`config`: Modify this to match the path of the JupyterLab configuration file.

`WorkingDirectory`: The working directory that JupyterLab opens to upon startup; you may change this as you see fit (it is recommended to set this to the user's home directory).

```
To check the JupyterLab installation path: which jupyter-lab
For the configuration file path: Refer to the path generated in the section
above.
```



4.2. Setting Up the Autostart Service

Enable the service to start automatically at boot:

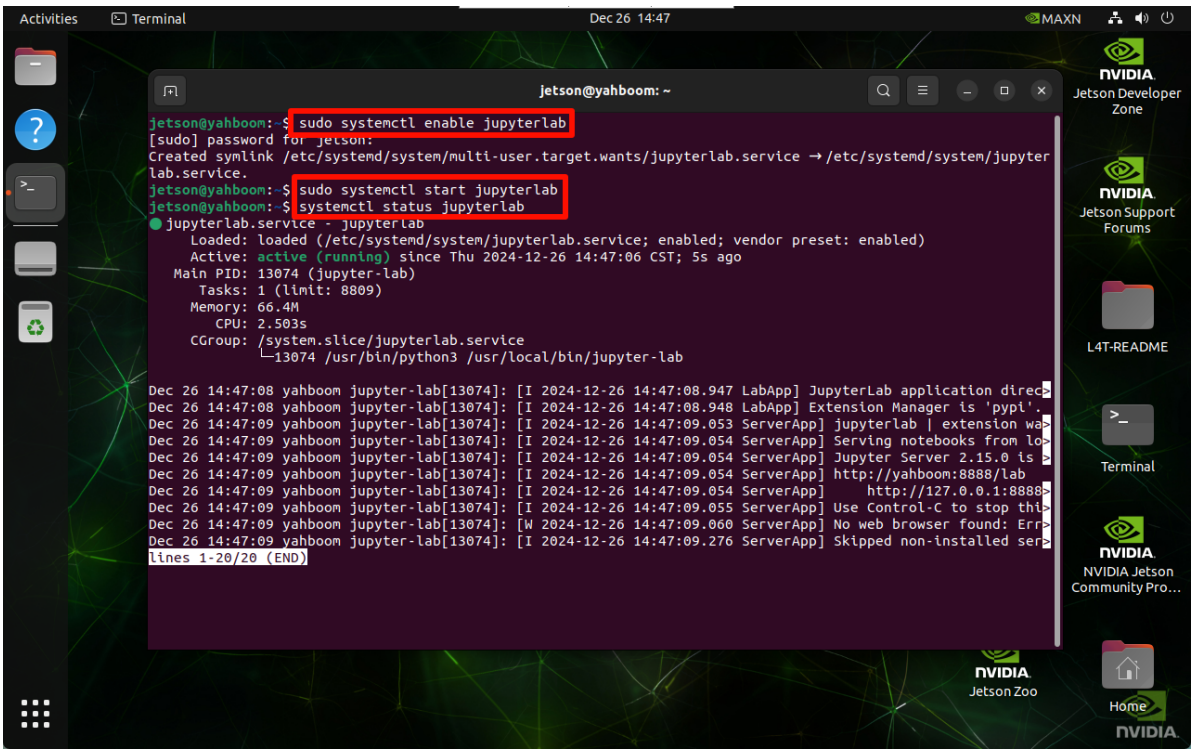
```
sudo systemctl enable jupyterlab
# To disable autostart: systemctl disable jupyterlab
```

Start the service:

```
sudo systemctl start jupyterlab
# Stop the service: sudo systemctl stop jupyterlab
```

Check the service status:

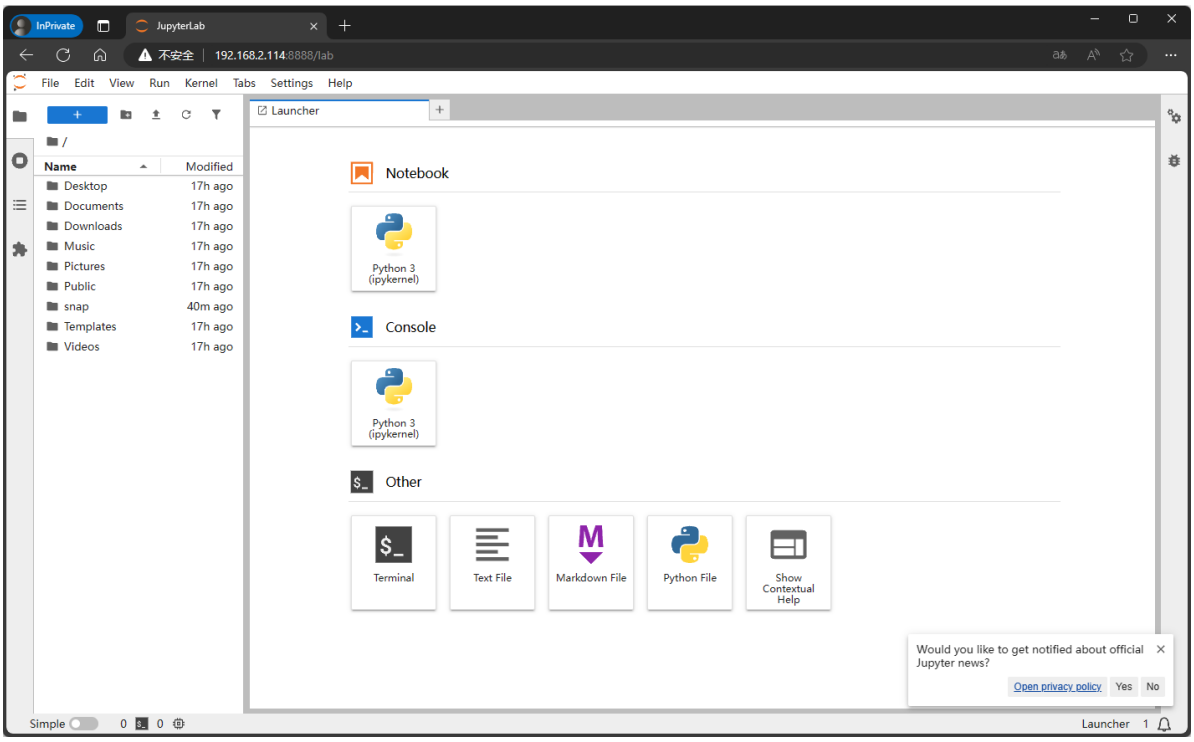
```
systemctl status jupyterlab
```



4.3. Verify Automatic Startup

After restarting the system, use a device on the same local area network (LAN) to access the board's IP address followed by port 8888 (e.g., `[Board_IP]:8888`).

```
Upon the first access, you will be prompted to enter a password. This password
corresponds to the credentials configured in the previous steps; for the factory
default image, the default password is: yahboom.
At the time these screenshots were taken, the board's IP address was
192.168.2.114; therefore, devices on the same LAN could access it via
192.168.2.114:8888.
```

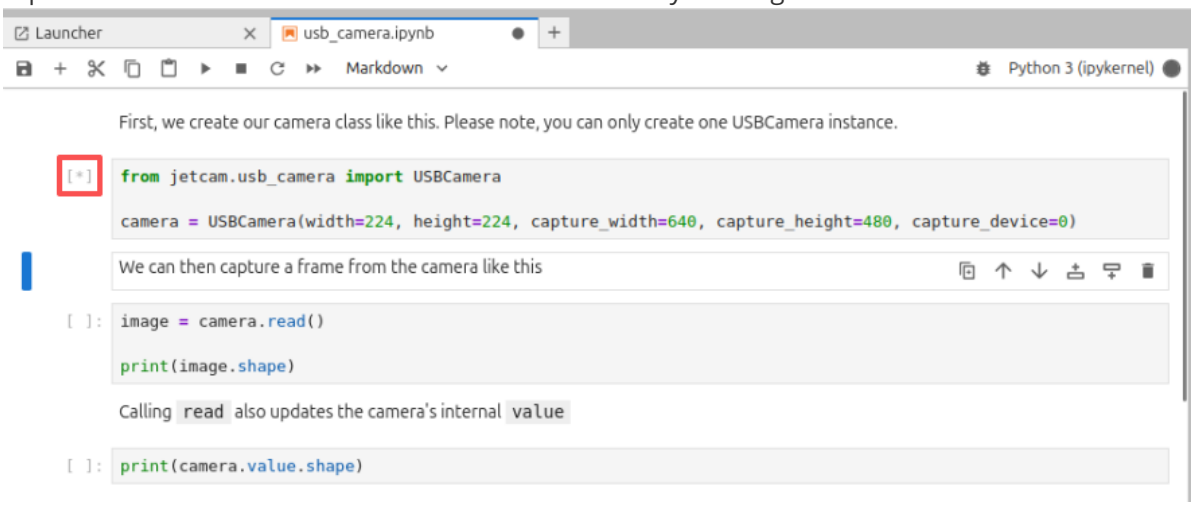


5. Using JupyterLab

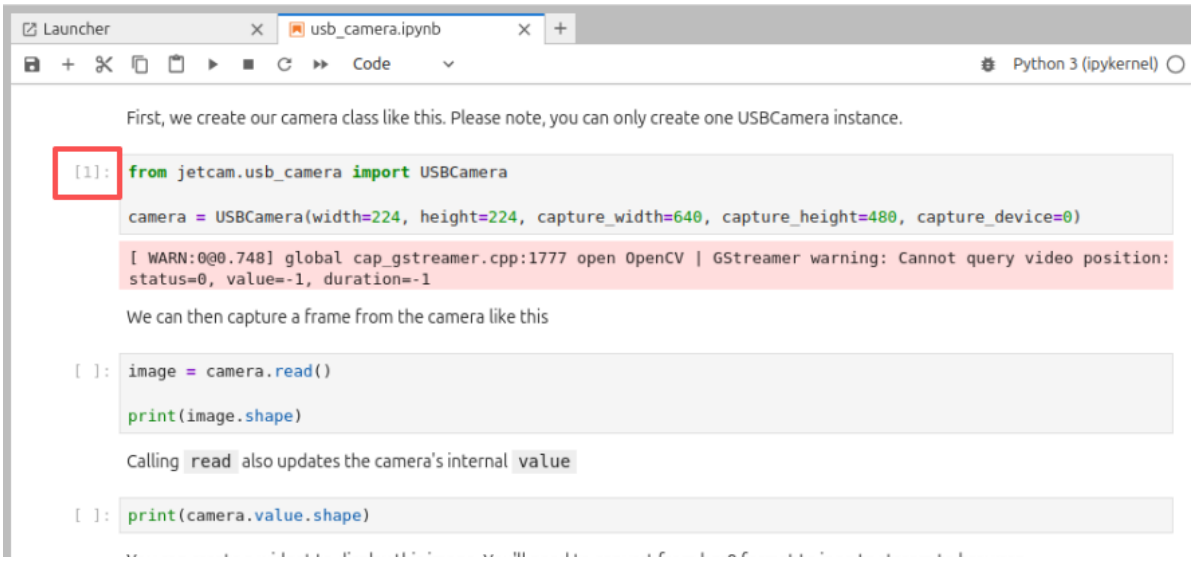
5.1. Running Programs

Use JupyterLab to open the program file you wish to execute.

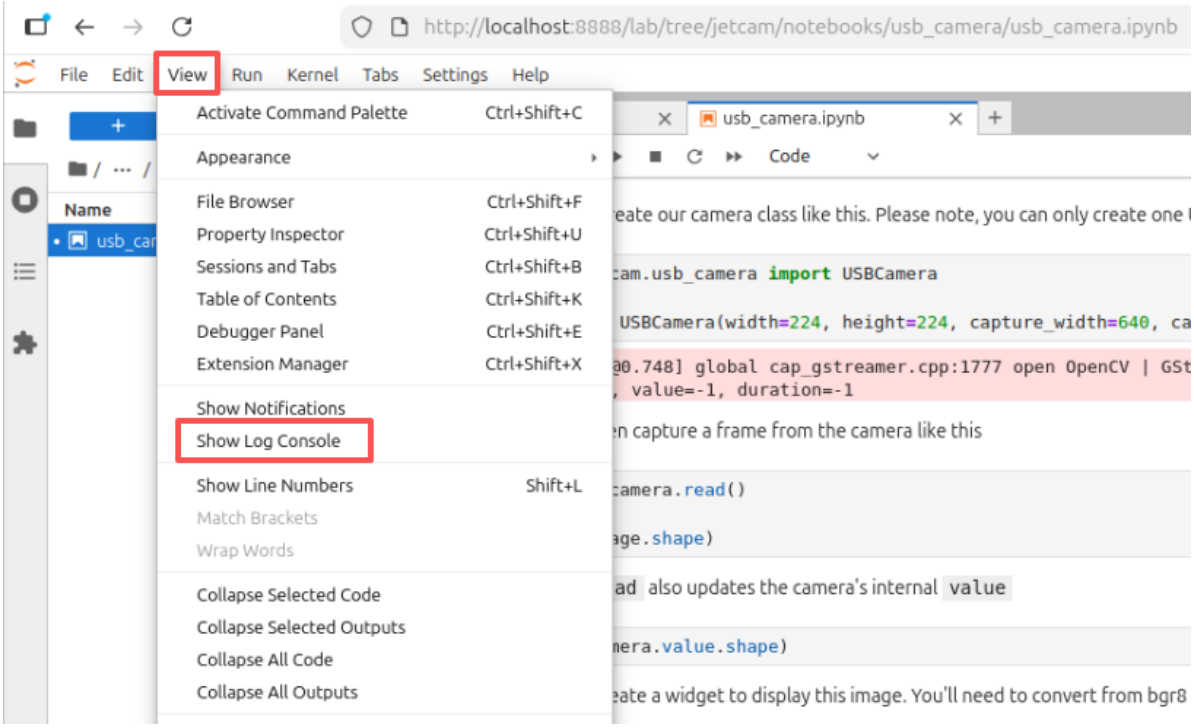
The JupyterLab toolbar is located above the program code. Hovering your mouse cursor over each menu item in the toolbar will display a tooltip describing its function; simply follow these prompts to perform the desired actions. From left to right, the toolbar functions are: 1. Save File; 2. Insert Cell Below; 3. Cut Selected Cells; 4. Copy Selected Cells; 5. Paste Cells; 6. Run Selected Cells; 7. Interrupt Kernel; 8. Restart Kernel; 9. Run All Cells; 10. Cell Type. An [*] displayed in the top-left corner of a cell block indicates that it is currently running:



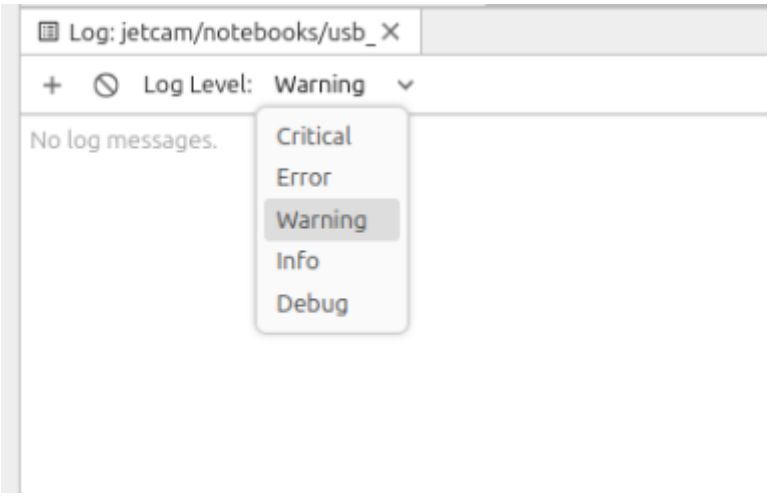
A [number] displayed in the top-left corner of a cell block indicates its sequential execution order; for example, [1] signifies that the program executed the code in this cell for the first time.



Click [View] -> [Show Log Console] to open the program debugging information panel.

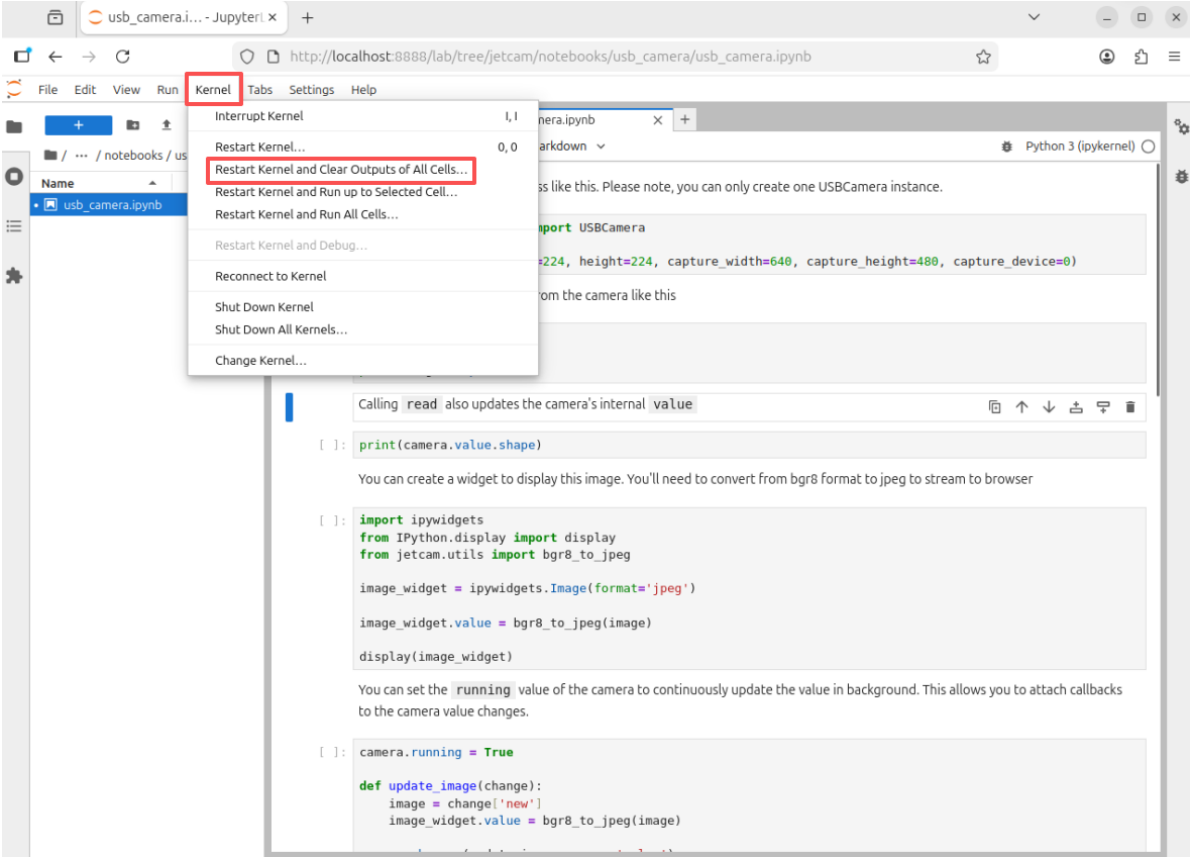


In the program debugging information window located at the bottom, you can adjust the message output level.

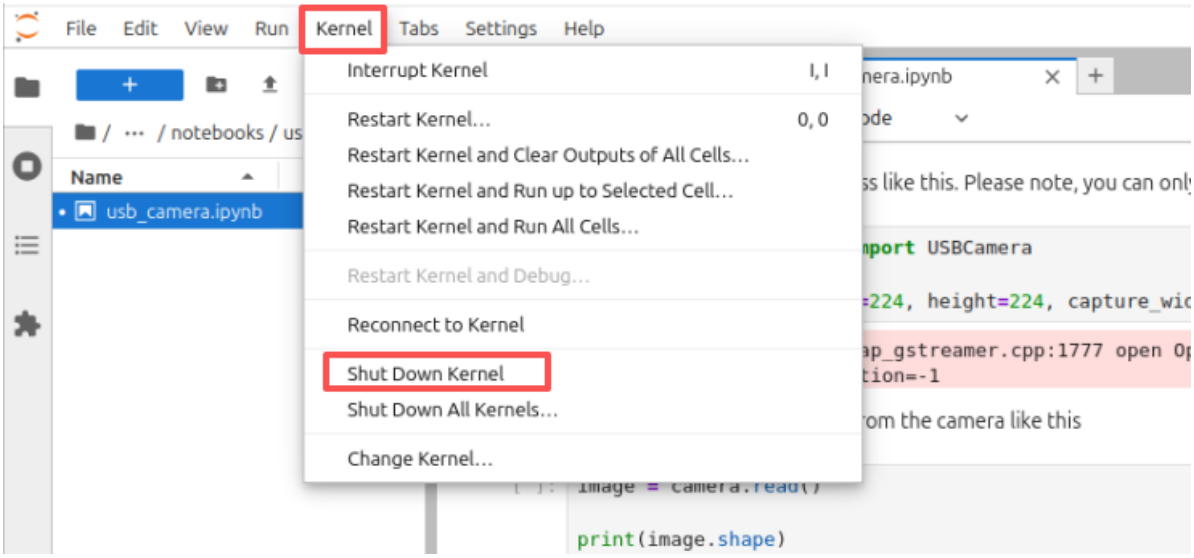


5.2. Kernel

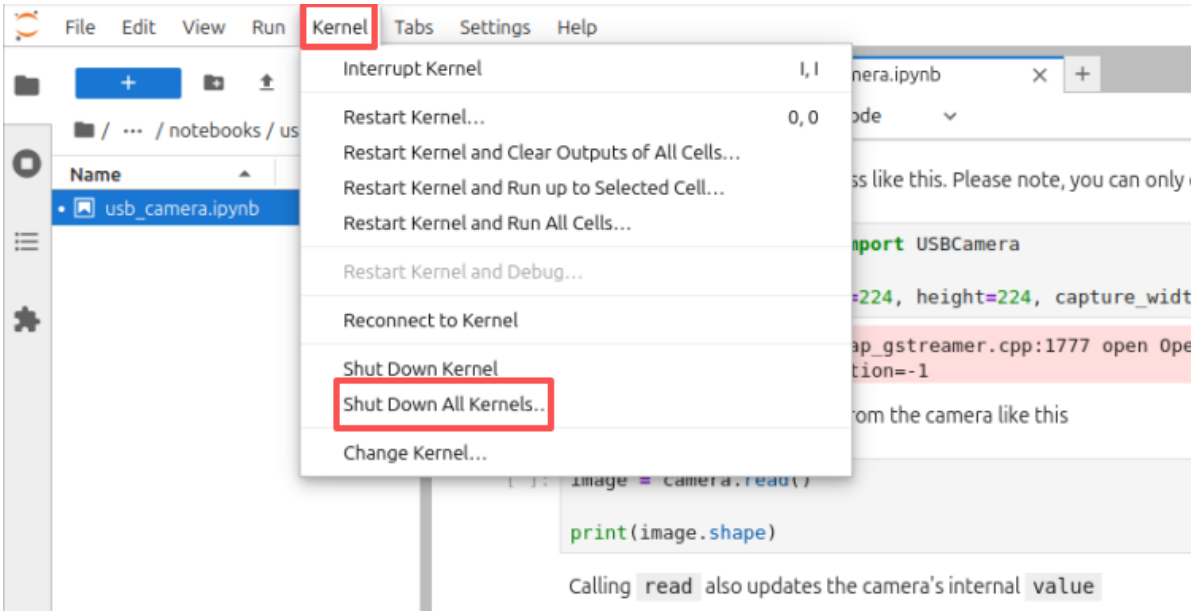
Restart Kernel: Click [Kernel] -> [Restart Kernel and Clear Outputs of All Cells]. It is recommended to restart the kernel and clear the output of all cells whenever you run the program or if the program encounters an error:



Shut Down Kernel: Click [Kernel] -> [Shut Down Kernel]. It is recommended to shut down the kernel promptly after you have finished using the program to prevent unnecessary resource consumption.



Shut Down All Kernels: Click [Kernel] -> [Shut Down All Kernels]. If other kernels are consuming system resources, you can shut down all kernels first, and then restart the program's specific kernel.


File
Edit
View
Run
Kernel
Tabs
Settings
Help
Interrupt Kernel
I, I
Restart Kernel...
0, 0
Restart Kernel and Clear Outputs of All Cells...
Restart Kernel and Run up to Selected Cell...
Restart Kernel and Run All Cells...
Restart Kernel and Debug...
Reconnect to Kernel
Shut Down Kernel
Shut Down All Kernels..
Change Kernel...
Name
usb_camera.ipynb
print(image.shape)
Calling read also updates the camera's internal value